

Task Manager API with PDF Export

Role: Software Developer

Difficulty: junior

Project Type: fullstack

Duration: 14 days

Focus: api_development

Candidate: Rahul Sharma

Project Overview

Build a RESTful API and simple UI for managing tasks, with JWT auth, role-based access, and PDF export of task lists.

Technologies

- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express.js
- Laravel (PHP)
- Tailwind CSS
- MongoDB
- JWT

Requirements

- Use existing skills, add new learning in API design and PDF export
- Implement role-based auth
- Create measurable endpoints

Objectives

- Implement auth flow
- Create CRUD for tasks
- Add PDF export
- Secure API
- Document endpoints

Functional Requirements

- User register/login
- Create/read/update/delete tasks
- Assign tasks to users
- Export tasks to PDF
- Search & paginate tasks

Non-Functional Requirements

- JWT auth
- Input validation
- Rate limiting
- Responsive UI
- Error handling
- Performance

Modules

Auth

User registration and login

- Register endpoint
- Login endpoint
- JWT issuance

Tasks

Task CRUD and assignment

- Create task
- Read task list
- Update task
- Delete task
- Assign user

Reports

PDF export of tasks

- Generate PDF
- Download endpoint

Admin

User role management

- Assign roles
- View users

Frontend

React UI

- Login page
- Task dashboard
- PDF download button

Backend

Express API

- Route handling
- Middleware
- Error handling

Implementation Plan

Setup

- Initialize repo
- Create backend & frontend folders
- Set up TypeScript

Auth

- Create user model
- Register & login routes
- JWT middleware

Tasks

- Task model
- CRUD routes
- Assign logic

Reports

- Integrate PDF library
- Create export route

Frontend

- Login form
- Task list UI
- PDF button

Testing & Docs

- Write unit tests
- Generate API docs
- Prepare README

Evaluation Criteria

API Design (25%)

RESTful, consistent endpoints

Security (20%)

JWT, input validation

Functionality (20%)

All features work

Code Quality (15%)

Clean, modular code

Testing (10%)

Unit & integration tests

Documentation (10%)

Clear README & API docs

Deliverables

- Source code
- README
- API docs
- Test report
- Sample PDF
- Deployment guide

Suggested Folder Structure

- frontend
- backend
- docs
- tests
- scripts

API Endpoints

POST /api/auth/register

Create user

POST /api/auth/login

Login user

GET /api/tasks

List tasks

POST /api/tasks

Create task

GET /api/tasks/:id

Get task

PUT /api/tasks/:id

Update task

DELETE /api/tasks/:id

Delete task

GET /api/reports/tasks

Export tasks PDF

Database Design

users

Store user data

- _id
- email
- passwordHash
- role

tasks

Store tasks

- _id
- title
- description
- status
- assignedTo
- createdBy
- createdAt

reports

Track exports

- _id
- userId
- generatedAt
- filePath

Testing Plan

- Unit tests for auth & tasks
- Integration tests for API endpoints
- E2E tests for login & PDF download

Candidate Fit

Rahul Sharma